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Decision: Reject

Submission: Combining Technical Indicators and Genetic Algorithms for Short-Term Machine Learning Prediction of Mini-Index Futures

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Summary of Reviews of [ijcnn_pap2597s1](#)

Reviewer	rec ⓘ	conf ⓘ	oral ⓘ	bp ⓘ	indust ⓘ
Reviewer 1	reject (1)	somewhat confident (2)	oral (1)	No (0)	Yes (1)
Reviewer 2	weak reject (2)	confident (3)	poster (0)	No (0)	No (0)
Reviewer 3	weak reject (2)	somewhat confident (2)	poster (0)	No (0)	No (0)
Reviewer 4	weak reject (2)	somewhat confident (2)	poster (0)	No (0)	No (0)
Averages:	1.8	2.2	0.2	0.0	0.2
Meta-Review	reject (1)				

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Full Reviews

Reviewer 1

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► Novelty and Significance

monospace

The paper presents a pipeline for short-term intraday futures prediction. However, the novelty and overall significance are limited. The approach relies on well-established techniques, including technical indicators, Information Gain-based feature selection, Genetic Algorithm hyperparameter optimization, and standard classifiers. Similar methodologies have been widely explored in prior work. As a result, the contribution is largely incremental and application-driven, with limited relevance to recent advances in financial machine learning.

► Relevance

monospace

The paper is generally relevant to the conference, as it addresses short-term financial time series prediction using machine learning methods. The topic aligns with applied neural networks and data-driven modeling. However, the work focuses on a conventional application with established techniques and does not strongly

connect to the broader or more recent research directions typically emphasized at the conference. As a result, its relevance is moderate rather than strong.

► Rigour and Soundness

markdown monospace

Weaknesses / Major Concerns

1. The methodology relies on relatively outdated techniques, including Genetic Algorithms and standard short-term machine learning models, and the manuscript lacks sufficient citation of recent related work.
2. The contribution is largely incremental, as it mainly combines technical indicators with Genetic Algorithm-based optimization without introducing clear methodological novelty.
3. No ablation studies are provided, making it difficult to assess the contribution of individual components.
4. The train/test evaluation protocol is not clearly time-respecting, raising concerns about potential data leakage in a non-stationary time-series setting.
5. The labeling procedure based on top and trough detection is insufficiently specified, making it difficult to assess lookahead bias and the effective prediction horizon.
6. The experimental evaluation uses limited baselines and omits stronger or more common tabular time-series models and time-aware validation schemes.
7. The writing and organization need improvement, particularly in the introduction, which lacks a clear motivation and discussion of limitations in existing approaches.

► Structure and Readability

monospace

The paper's structure and organization need significant improvement and do not fully meet standard academic writing conventions. The introduction lacks a clear research motivation and does not adequately position the work with respect to the limitations of existing methods. The cited literature is relatively outdated, which further weakens the contextual grounding of the study. The methodology section is missing important algorithmic details, reducing clarity and reproducibility, and no ablation experiments are included to support the design choices. In addition, the subsection on Genetic Algorithm parameters would be more appropriate in the experimental setup rather than in the methodology section. Overall, substantial reorganization and clarification are needed to improve readability and coherence.

► Best Paper Justification

None

► Other Comments for Authors

None

Scored Review Questions

rec ⓘ

conf ⓘ

oral ⓘ

bp ⓘ

indust ⓘ

ijcnn_pap2597s1 reject (1) somewhat confident (2) oral (1) No (0) Yes (1)

Reviewer 2

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► Novelty and Significance

This paper offers a limited contribution in terms of novelty. The study's core approach involves applying a mature optimization technique—the Genetic Algorithm (GA)—to a standard financial forecasting workflow that includes feature selection and hyperparameter tuning. While the process is complete, this combination of techniques is relatively common in the fields of machine learning and computational finance. The paper does not propose methodological innovations or offer profound new insights based on this application. More critically, the significance of the research is not sufficiently demonstrated. In an application-driven field like financial forecasting, the value of a study must ultimately be measured by its economic significance. The evaluation in this paper is focused entirely on predictive accuracy and F1-score, overlooking key indicators of economic viability such as strategy backtesting, risk metrics (e.g., Sharpe ratio, maximum drawdown), and transaction costs. Therefore, the practical value of the reported performance improvement (from approx. 61% to 65%) is difficult to assess, leaving the significance of the research questionable.

► Relevance

monospace

The research topic of this paper—financial market forecasting using machine learning—is relevant to the fields of computational finance, data science, and artificial intelligence applications.

► Rigour and Soundness

monospace

The paper exhibits several significant deficiencies in its scientific rigour and soundness, which weakens the persuasiveness of its conclusions.

First, its core evaluation metric has limitations. In financial trading, classification accuracy by itself is an insufficient measure, as it fails to reflect a model's profitability, risk exposure, and real-world trading frictions. An evaluation that omits these economic dimensions is incomplete, making it impossible to ascertain the effectiveness of the research findings in a practical application.

Second, the experimental design has notable weaknesses. Comparing the optimized models solely against an "un-optimized" baseline model is a relatively weak benchmark. A more convincing experiment would include comparisons against other established or simpler alternative strategies in the domain. Furthermore, despite the use of cross-validation, performing both feature selection and intensive hyperparameter search on a relatively limited dataset significantly increases the risk of data overfitting. The large performance gap of the Random Forest model between cross-validation and the test set indirectly supports this concern.

► Structure and Readability

monospace

The paper is well-structured and fluently written, allowing readers to easily follow the technical workflow and experimental steps. The quality of the figures (such as the flowchart and convergence curves) is also good and effectively aids in presenting the content.

However, a good presentation cannot compensate for the core methodological shortcomings of the research. The paper's clarity, in fact, makes its limitations in evaluation metric selection and experimental design more apparent. While the manuscript is easy to read, its fundamental issues concerning rigour and soundness make it difficult to meet the standards required for academic publication.

► Best Paper Justification

None

► Other Comments for Authors

None

Scored Review Questions

rec ⓘ

conf ⓘ

oral ⓘ

bp ⓘ indust ⓘ

Reviewer 3

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► Novelty and Significance

monospace

This article investigates short-term trend forecasting in the mini-index futures market using a hybrid framework that integrates technical analysis, feature selection, and evolutionary hyperparameter optimization. From an application perspective, the novelty of this work lies in its utilization of Five-minute intraday data from the Brazilian Stock Exchange, particularly the WIN futures contracts, thereby providing a specific case study for emerging markets. However, from a methodological standpoint, the work merely combines feature selection and hyperparameter optimization within a unified evolutionary framework. The core techniques themselves (technical indicators, feature selection, and GA tuning for machine learning models) are derived from existing literature, thus lacking any breakthrough innovation.

► Relevance

monospace

This paper is highly relevant to IJCNN. IJCNN emphasizes neural networks and computational intelligence, and this work employs Multilayer Perceptron, Support Vector Machine, Random Forest, and Genetic Algorithms for optimization—all of which are intrinsically linked to neural computation. The related work section references applications of MLP in financial forecasting and extends these to comparisons with other models (such as RF and SVM). Its focus on short-term forecasting in financial markets further illustrates the real-world utility of computational intelligence.

► Rigour and Soundness

monospace

The rigour and soundness of the paper are moderately below average. While the overall technical framework is reasonable, there are notable shortcomings in both the methodology and experimental aspects that compromise its rigor. In the methodology section, the feature selection process merely states that Information Gain Method outperforms other methods, without providing comparative data or detailed justifications for its selection; similarly, the genetic algorithm parameters are fixed (e.g., population size of 10 and 1000 generations) without explanation of why these settings are appropriate or any sensitivity analysis to demonstrate robustness. In the experimental section, the study relies solely on an untuned Random Forest as the baseline, which could be significantly strengthened by comparisons with alternative tuning methods or more complex model architectures. Furthermore, the absence of benchmarks involving non-evolutionary algorithms makes it difficult to independently evaluate the actual contribution of the genetic algorithm.

► Structure and Readability

monospace

The overall structure of the paper is reasonable. However, the clarity of the specific content is somewhat lacking. The methodological section largely states what was done but omits crucial implementation details (e.g., pseudo-code, internal module diagrams, and explicit justifications for chosen parameter ranges), making it difficult for readers to understand or reproduce the approach.

► Best Paper Justification

None

► Other Comments for Authors

None

Scored Review Questions

rec ⓘ

conf ⓘ

oral ⓘ

bp ⓘ

indust ⓘ

ijcnn_pap2597s1 weak reject (2) somewhat confident (2) poster (0) No (0) No (0)

Reviewer 4

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► Novelty and Significance

monospace

The manuscript explores a hybrid machine learning framework for predicting 5-minute intraday trends in the Brazilian Mini-Index futures market by combining Information Gain-based feature selection with Genetic Algorithm (GA) hyperparameter optimization. While the integration of evolutionary algorithms with classifiers like SVM and MLP is a known methodology in financial forecasting, the study provides specific empirical value by applying these techniques to the BovDbV2 dataset and the high-liquidity WIN contracts. The primary significance lies in the systematic demonstration that joint optimization yields a measurable performance boost, elevating test accuracy from a 61.63% baseline to a peak of 65.85% for the Multilayer Perceptron

► Relevance

monospace

relevant

► Rigour and Soundness

monospace

The experimental design is rigorous, utilizing a balanced training set of 3,000 instances to prevent class bias and a large, unseen test set of 12,057 instances to validate generalization. The methodology correctly applies 5-fold cross-validation during the optimization phase and maintains consistency across all models by using the same GA configuration and a reduced feature set of seven key indicators. However, the results reveal a significant generalization gap for the Random Forest model, which achieved a cross-validation fitness of 0.7739 but dropped to 0.6331 on the test set. While the authors attribute the superior performance of SVM and MLP to their ability to capture non-linear dynamics and margin-based learning, a more technical critique of why the GA-optimized RF failed to generalize compared to its counterparts would further strengthen the scientific rigor.

► Structure and Readability

monospace

The paper is well-structured and follows a standard technical progression, effectively utilizing visual aids to document the evolutionary process. Figure 1 provides an essential overview of the methodological flow, while Figures 2 through 4 clearly illustrate the convergence of the fitness function over 1,000 generations. The inclusion of detailed tables (Tables I, II, and III) listing the specific search spaces and optimal hyperparameters—such as the L_2 regularization of 4.48×10^{-6} for the MLP—ensures high reproducibility. The comparison between the RMSprop and Adam optimizers in Table IV adds a layer of depth that clarifies how different training algorithms influence the precision-recall trade-off within the same GA-optimized search space

► Best Paper Justification

None

► Other Comments for Authors

None

Scored Review Questions

rec ⓘ

conf ⓘ

oral ⓘ

bp ⓘ

indust ⓘ

ijcnn_pap2597s1 weak reject (2) somewhat confident (2) poster (0) No (0) No (0)

Meta-Review

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Meta-Review

monospace

This study presents a financial market forecasting workflow using Genetic Algorithms. As the reviewers pointed out, the work focuses on a conventional application with established techniques and does not strongly connect to the broader or more recent research directions. The research has moderate relevance to this conference. Its contribution is limited in terms of novelty, significance, and experimental design. Considering that all the reviewers recommend rejection consistently, it would be reasonable to reject this work.

Scored Questions

Overall recommendation *reject*

Committee Comments

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Committee Comments for Authors

None